γ-Ray Spectroscopy with HpGe(Li) Detector

Student 1

Ohio University,
Department of Physics and Astronomy, Athens, OH, 45701, USA
Email:: #######@ohio.edu
April 10, 2015

Abstract. The main propose of this experiment is to measure the abundance ratio of isotopes in a sample of Uranium Ore, which is also called Yellow cake. The HpGe detector is calibrated using ^{22}Na, ^{60}Co and ^{152}Eu γ-ray spectra. The energy resolution of system and energy dependence of the detector absolute efficiency are also calculated. The isotopic abundance of Uranium ($^{235}U/^{238}U$) are $7.16 * 10^{-3}$, $2.004 * 10^{-3}$ and $3.62 * 10^{-2}$ for natural uranium, depleted uranium and enriched uranium respectably [1]. We have found this ratio be: $2.39 * 10^{-3} \pm 7.9 * 10^{-12}$ for the depleted one which deviates from the literature value by about 19%. This indicates that the concentration of ^{235}U in Yellow Cake is very less than that of ^{238}U. Due to the experimental miss conduct for metallic Uranium sample, we did not compute ($^{235}U/^{238}U$) ratio for Metallic Uranium sample.

Keywords: γ-ray spectroscopy, HpGe detector, activity, disintegration

1 Introduction

Gamma-Ray Spectroscopy is to tool to study energy spectra of γ-rays. γ-rays are electromagnetic radiations of highest energies. The natural range for γ-rays emitted by nuclei is 0.1 20 MeV. Gamma rays are emitted in nuclear processes during radioactive decay or in nuclear reactions. When a nucleus in excited energy level, goes back to ground level, it releases energy in the form of gamma ray photon. Typically, nuclei de-excite after a nuclear reaction by emitting a few nucleons (which removes much of the energy) and then by γ-ray cascade which removes most of the angular-momentum. In order to isolate the interesting cascades, two γ-rays close in energy should be distinguishable. This means that the peak associated with γ-rays of a given energy in a spectrum should be as narrow as possible. This characteristic is called high energy-resolution. This is fulfilled in Germanium detector because of the small energy gap between the valence and the conduction band. In this experiment, the isotope ratio ($^{235}U/^{238}U$) is going to be determined using the Germanium detector and result will be compared with literature values [1].

1.1 Theoretical Background

Gamma ray photons are emitted in nuclear processes during radioactive decay or in nuclear reactions with energy ranging from 0.1 Mev to 20 Mev range. These gamma rays interact with matter and lose energy through the following ways [3]:

- **Photoelectric Effect**- Gamma ray photon interact with detector atom in which the photon completely disappears and an energetic photo-electron is ejected by the atom from one of its bound shells. This interaction is only with the atom as a whole not with free electrons.The energy of the photo-electron is equal to the difference of energy of the incoming photon and work function of the electron in the atom.
- **Compton Scattering**-In this process, a portion of photon energy is transferred to the electron and is deflected at some angle with respect to its original direction. The electron recoils and is called recoil electron. Because all angles of scattering are possible, the energy transferred to the electron can vary from zero to large fractions of gamma photon energy.
- **Pair Production**- Pair production is energetically possible for gamma rays with energies above 1.02 Mev, i.e the sum of rest mass energies of electron and positron. In presence of coulomb field of the nucleus, the gamma ray photon disappears and is replaced by a positron-electron pair. Practically the probability of pair production is very low until energy of gamma rays is several Mev.

Germinium Detector:

High-Purity Germanium (HPGe) detector is a semiconductor particle detector and is used in γ-ray spectroscopy. HPGe is a semiconductor diodes having a p-n structure. These detectors directly collect the charges produced when the semiconductor material is ionized by the ionizing radiations. Electron-hole pairs are produced by ionization. These pairs drift under an external electric field to +ve and -ve electrodes where they generate the pulse. Germanium has relatively low band gap of 0.67 eV at room temperature, so the detector must be cooled in order to reduce the thermal generation of charge carriers. Otherwise, leakage current destroys the energy resolution of the detector. Liquid nitrogen, which has a temperature of -120^0 is used to cool the detector[2]. The schematic diagram of p-n diode is as shown in Fig.1.

Relative photo peak efficiency is calculated using following Eq.11.

$$Relative\ efficiency = \frac{Total\ area\ under\ peak}{Intensity\ of\ the\ source\ peak} * 100 \tag{1}$$

And Absolute efficiency is calculated as:

$$Absolute\ efficiency = \frac{R_1}{R_2} * 100 \tag{2}$$

Where R_1 is detected emission rate given by Eq.3,

$$R_1 = \frac{\sum N}{t} \tag{3}$$

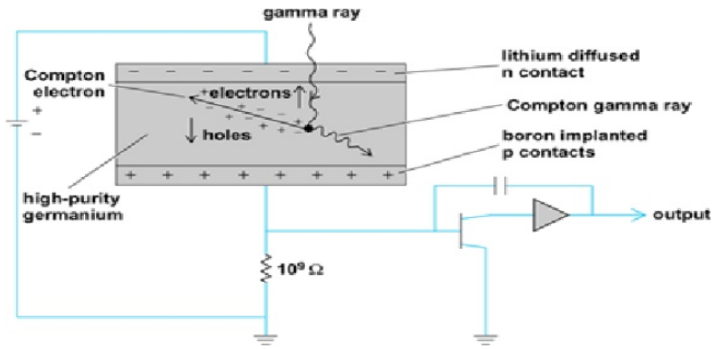


Fig. 1. Configuration of p-n diode HpGe Detector [5]

N is the total number of counts of given photo peak and t is the time taken to take the data. R_2 is given by the Eq.4

$$R_2 = N_0\ e^{-\lambda t} \tag{4}$$

Where t is time in days from May 1, 1983 to Feb 10, 2015, λ is decay constant and N_0 is the emission rate at time t = 0 second.
The number of Uranium nuclei in the sample can be calculated as:

$$\frac{dN}{dt} = -\lambda N \tag{5}$$

The solution of Eq.5 is:

$$N = N_0\ e^{-\lambda t} \tag{6}$$

Where N is the number of nuclei decayed at time time t and λ is the decay constant.

$$\frac{dN}{dt} = -\lambda N = \frac{DR}{BF} = \frac{measured\ decay\ rate\ R_1}{Branching\ factor} \tag{7}$$

So,

$$N = \frac{DR}{BF} * \frac{1}{\lambda_{total} * \epsilon} \tag{8}$$

Where $\lambda_{total} = ln(2)/t_{\frac{1}{2}}$ and ϵ is efficiency.

2 Materials and Experimental Procedures

2.1 Laboratory Equipment

This figure 2 shows a block diagram of the experimental setup of HPGe detector. γ-ray from source falls on the germanium crystal in the detector. The energy of

the photon is transferred to atoms in the crystal and electrons are ejected. This forms electron-hole pairs which forms a pulse. So the detector gives the signal in form of current. This signal not being so strong is connected to a preamplifier whose function is to amplify signal. the signal from preamplifier goes to amplifier whose function is to further amply and to change -ve polarity to +ve polarity of signal. Amplifier is connected to the Analog to digital converter (ADC) to convert signal to particular digital form (channel #), in which the channel # is directly proportional to the photon energy.

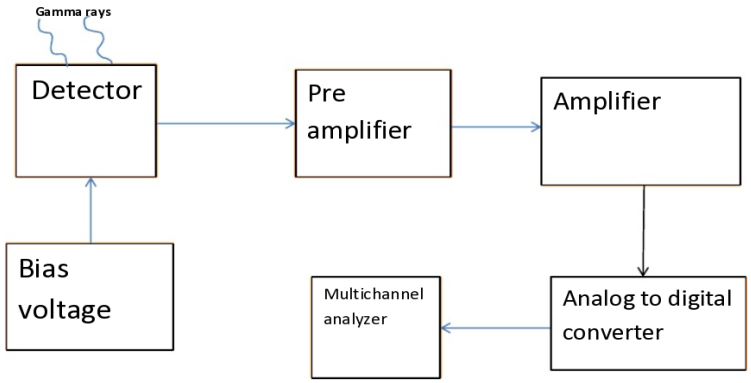


Fig. 2. Electronic set up for HpGe Detector [5]

We have used ^{22}Na,^{60}Co,^{152}Eu, and mixture of^{235}U and ^{238}U (also called Yellow cake) in our experiment.

2.2 Experimental Procedures

First of all, electronic connections were made. The liquid nitrogen to the detector Dewar was checked and it was already full. The bias voltage to the detector was increased to 2500 V. The preliminary data was taken to set the coarse gain, fine gain and shaping time for the desired range of the spectra. The coarse gain was set to 50 with fine gain 0.71 with shaping time 1μ sec. The background spectra was taken without sample for 10 minutes and saved in the disk 2005 at tag# 0. The samples in this experiments were ^{22}Na,^{60}Co,^{152}Eu, and mixture of^{235}U and ^{238}U (Yellow cake). Then, the spectra of above samples were taken for 10 minutes each and saved in the same disk.

3 Results and Discussion

3.1 Calibration of Detector

^{22}Na,^{60}Na and ^{152}Eu were used to calibrate HPGe detector. The spectra of background and background subtracted spectra of ^{22}Na,^{60}Na, ^{152}Eu and Uranium with Gaussian fit to each peaks are shown below:

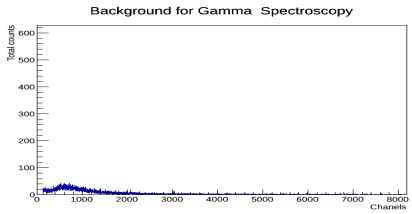


Fig. 3. Background spectra

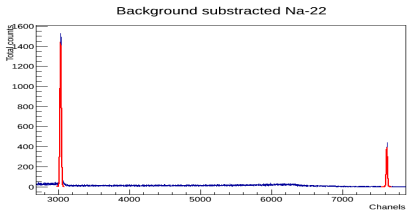


Fig. 4. ^{22}Na without background

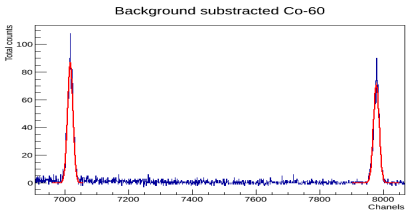


Fig. 5. ^{60}Co without background

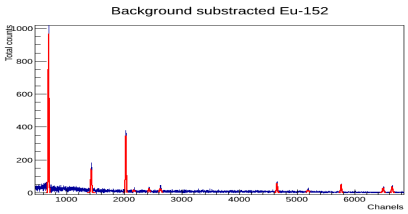


Fig. 6. ^{152}Eu without background

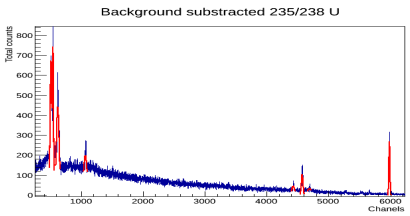


Fig. 7. Metallic Uranium spectra with Gaussian fit

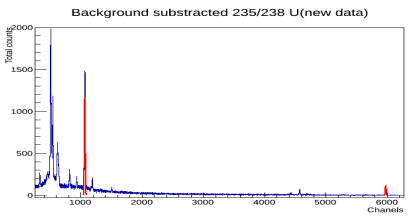


Fig. 8. Uranium (Yellow cake) spectra (new data taken)

Gaussian fit was performed for each peak to find the amplitude, mean channel number, standard deviation and number of counts and their corresponding errors. The results are presented in the following Table.1.

photopeak	Amp.	channel#	channel err.	std	std error	counts
peak 1 of co-60	87.21	7017.85	0.199	8.535	0.16	1794 ± 45
peak 2 of co-60	70.90	7976.58	0.230	8.974	0.18	1653 ± 41
peak 1 of Na-22	1409.73	3030.87	0.053	9.85	0.05	34407 ± 191
peak 2 of Na-22	930.84	7628.47	0.095	8.74	0.08	8777 ± 94
peak 1 of Eu-152	967.24	685.32	0.052	6.93	0.052	15762 ± 135
peak 2 of Eu-152	136.59	1426.01	0.1 66	8.99	0.206	1785 ± 57
peak 3 of Eu-152	343.39	2026.26	0.091	7.12	0.079	5776 ± 81

Table 1. The table shows Gaussian fit parameters with their errors and counts

$$Energy = slope * channel + intercept \; and \; Error = \frac{SD}{\sqrt{N}} \tag{9}$$

$$Calculated \; Energy = slope * channel + intercept \tag{10}$$

Slope	Slope Error	Intercept	Intercept Error
0.166	$2.305 * 10^{-6}$	7.9102 KeV	0.0082 KeV

Table 2. linear fit parameters

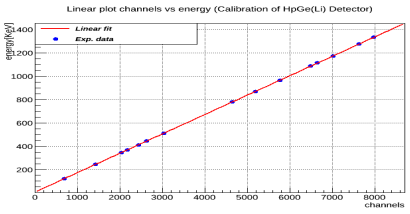


Fig. 9. Linear fit, the energy calibration of detector

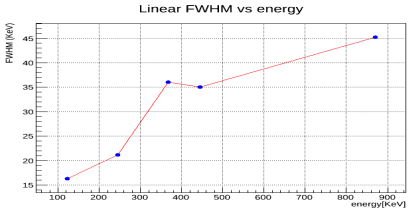


Fig. 10. Plot of fwhm vs calculated energy

We have plotted the fwhm with respected to the calculated energy of five peaks of ^{152}Eu in the Fig.10. This shows that the FWHM is increased with increasing energy values of peaks.Calibration curve (Fig.9) was plotted by using the photo-peak energies of ^{22}Na,^{60}Na and ^{152}Eu . Energy associated with each channel can be calculated by using Eq.10. Error in channel is given by Eq.9. The linear fit parameters and their errors are tabulated in the Table 2. Now, the calculated energy of corresponding peaks are calculated using Eq. 10.

The resolution energy, calculated energy, expected energy, residual energy and peak errors of peaks of corresponding spectra are tabulated in the Table

Photopeaks	Exp. Energy [4] (KeV)	Cal. energy (KeV)	Res. Energy (KeV)	Resolution (KeV)	Peak Err. KeV
peak1 of ^{22}Na	511	511.093	0.093	0.045	0.008
peak2 of ^{22}Na	1274.54	1274.38	-0.153	0.0161	0.013
peak1 of ^{60}Co	1173.23	1173.01	-0.2200	0.0171	0.026
peak2 of ^{60}Co	1332.49	1332.18	-0.3150	0.0158	0.029
peak1 of ^{152}Eu	121.782	121.688	-0.0943	0.1339	0.009
peak2 of ^{152}Eu	244.697	244.655	-0.0417	0.0863	0.034
peak3 of ^{152}Eu	344.279	344.309	0.0302	0.0486	0.013
peak4 of ^{152}Eu	367.789	367.547	-0.242	0.0981	0.021
peak5 of ^{152}Eu	411.116	411.323	0.2070	0.0666	0.087
peak6 of ^{152}Eu	443.965	444.032	0.0670	0.0789	0.139
peak7 of ^{152}Eu	778.904	779.85	0.9460	0.0265	0.039
peak8 of ^{152}Eu	867.373	869.282	1.9090	0.0520	0.023
peak9 of ^{152}Eu	964.079	964.652	0.5730	0.0206	0.042
peak10 of ^{152}Eu	1085.87	1085.95	0.0840	0.0237	0.076
peak11 of ^{152}Eu	1112.07	1111.82	-0.2450	0.0219	0.053

Table 3. Expected Energy, Calculated Energy, and Residual Energy, resolution and Peak Error

3. From Table3, it is seen that the peak errors in energies are smaller than resolution energies of the detector for almost all the given peaks.

3.2 Efficiency of HPGe Detector

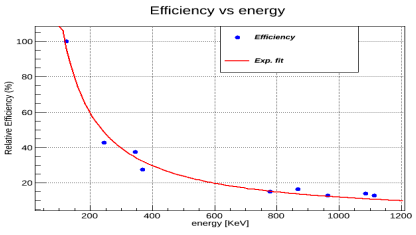


Fig. 11. Efficiency plot of HPGe detector

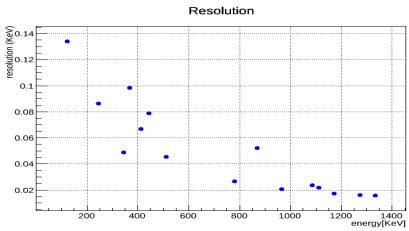


Fig. 12. Resolution of HpGe detector

We have used ^{152}Eu source to find the efficiency of the given HPGe detector. The necessary informational parameters to find the efficiency of detector are tabulated in the Table4. The intensity ratios for both counts and intensity for lowest energy peak is taken as 100% and count ratio and intensity ratio for other peaks are compared with that peak, which is shown in the Table4.

We have plotted the relative efficiency(%) vs energy (KeV) of corresponding peaks of ^{152}Eu in the Fig.11. From the Fig.11, it is seen that the relative

Photopeaks	Lit. Energy (KeV)	Cal. energy (KeV)	Counts	count ratio(%)	Int(%)	Int. ratio(%)	Efficiency ϵ(%)
peak1 of ^{152}Eu	121.782	121.688	15762	100	28.53	100	100
peak2 of ^{152}Eu	244.697	244.655	1785	11.32	7.55	26.46	42.78
peak3 of ^{152}Eu	344.279	344.309	5776	36.65	26.59	93.20	39.32
peak4 of ^{152}Eu	367.789	367.547	130	0.82	0.859	3.01	27.40
peak5 of ^{152}Eu	411.116	411.323	578	3.67	2.237	7.84	46.81
peak6 of ^{152}Eu	443.965	444.032	699	4.44	2.827	9.90	44.84
peak7 of ^{152}Eu	778.904	779.85	1077	6.83	12.93	45.32	15.07
peak8 of ^{152}Eu	867.373	869.282	382	2.42	4.23	14.83	16.32
peak9 of ^{152}Eu	964.079	964.652	1032	6.55	14.51	50.86	12.88
peak10 of ^{152}Eu	1085.87	1085.95	777	4.93	10.11	35.44	13.91
peak11 of ^{152}Eu	1112.07	1111.82	950	6.03	13.67	47.91	12.58

Table 4. Plot with relative efficiency

efficiency of the HPGe detector is decreased exponentially and reaches less than 10% beyond 1000 KeV energy..

3.3 235/238 Uranium Isotope Abundance in the "Yellow Cake"

Uranium Ore ("Yellow Cake") is a mixture of ^{235}U and ^{238}U. Both are radioactive isotope of Uranium which are spontaneously disintegrate to their daughter nuclei producing γ-rays of different energies. The Uranium Ore decay spectra is plotted in the Fiq.7. We have detected eight peaks which correspond to different decay scheme of ^{235}U and ^{238}U. Extected energy, calculated energy, amplitudes of peaks, channel #, residual energy, fwhm, peak error and counts of observed eight peaks of Uranium Ore is tabulated in the Table.5. The table5 also shows the errors in counts of the different eight peaks of Uranium Ore spectra.The resolution energy of the detector is greater than the peak errors which implies that the peak errors in our data are not significant for the analysis.

Photo peaks	lit. en. KeV	Cal. En. KeV	Amp.	ch. #	Res. En. KeV	fwhm KeV	P. Er KeV	Counts
peak 1	93.35	93.064	672.807	512.917	-0.286	29.51	0.031	11334±138
peak 2	98.44	98.557	716.731	546.001	0.114	22.03	0.024	10635±129
peak 3	111.35	111.519	441.025	624.079	0.169	53.29	0.044	8709±151
peak 4	185.72	185.496	199.095	1069.67	-0.219	66.83	0.166	2672±102
peak 5	743.00	743.332	44.996	4429.74	0.332	52.19	0.193	867±47
peak 6	766.60	767.461	103.281	4575.07	0.861	31.67	0.059	2426±63
peak 7	786.40	786.984	33.453	4692.67	0.584	67.84	0.424	587±43
peak 8	10001.20	1001.540	268.291	5985.02	0.339	19.62	0.016	5559 ±77

Table 5. Expected Energy, Calculated Energy, amplitude channel #, Residual Energy, fwhm, Peak Error and counts of Uranium Yellow Ore

We have plotted the 8th peak and 14th peak of Uranium Ore spectra(new data taken) in the Fig.13 and Fig.14 which correspond to the nuclei ^{235}U and ^{238}U with calculated energies of 185.496 KeV and 1001.20 KeV respectively.

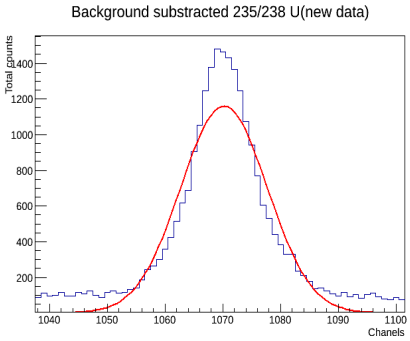


Fig. 13. Gaussian fit of 8th peak of Uranium Ore spectra(new data)

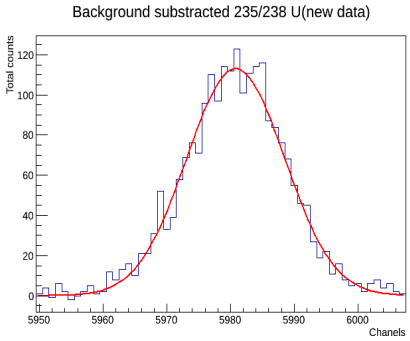


Fig. 14. Gaussian fit of 14th peak of Uranium Ore spectra (new data)

Photo peaks	Cal. En. KeV	Decay rate counts/sec	Branching Ratio[6](%)	$t_{1/2}$ (years)	rel. eff. ϵ(%)	^{235U}N & ^{238U}N from Equation 8
peak 8	185.496	30.90 ±0.26	58.077	$6.038(5) * 10^8$	67.0	$1.51 * 10^{14} \pm 7 * 10^5$
peak 14	1001.54	3.73±0.08	0.839	$4.468(3) * 10^9$	10.0	$6.26 * 10^{16} \pm 3 * 10^6$

Table 6. Calculation of ^{235}U and ^{238}U Number of counts

To find the 235/238 Uranium Isotope Abundance in the "Yellow Cake", we have tabulated necessary parameters in the Table 6.

Hence from the Table 6, $N_{235} = 1.51 * 10^{14} \pm 5 * 10^5$ and $N_{238} = 6.26 * 10^{16} \pm 3 * 10^6$. Hence, Isotope abundance ratio of Uranium Ore is given by::

$$\frac{N_{235}}{N_{238}} = \frac{1.51 * 10^{14} \pm 5 * 10^5}{6.26 * 10^{16} \pm 3 * 10^6} = 2.39 * 10^{-3} \pm 7.9 * 10^{-12} \tag{11}$$

This value of ratio corresponds to depleted Uranium Ore. This ratio indicates that the percentage of ^{235}U is very less in comparison to the ^{238}U in "Yellow Cake". The average value of this ratio is $\frac{N_{235}}{N_{238}} = 2.004 * 10^{-3}$ [1]. Our result is deviated by about 19%. In my opinion, this deviation may be due to the imperfect position of the sample with respect to the detector. Another cause of error may be due to the effective area of detection of HPGe detector, which we assume full surface area for the detection of γ-rays, which may not be true.

3.4 Conclusion

Calibration of HPGe detector done by using ^{22}Na,^{60}Na, and ^{152}Eu spectra. Corresponding peak energies of of the spectra were calculated using calibration parameters. The resolution was calculated and was found to be small (0.016 KeV) which is decreased with increasing values of energies. The full width at half maximum were calculated which increased with peak energy. The relative efficiency of the detector was calculated which is decreased exponentially with respect to the peak energies of the spectra. We have found that the relative efficiency of ^{235}U is more than ^{238}U which can be seen in the plot. The isotopic ratio of ^{235}U and ^{238}U in depleted Uranium Ore ("Yellow Cake") was found to be: $2.39 * 10^{-3} \pm 7.9 * 10^{-12}$ which is deviated by approx. 19% than that of literature value. This may be due to the defect of detection surface area of the HPGe detector.

Acknowledgment: I would like to thank Dr. Paul King for his guidance and support throughout this experiment.

References

1. http://www.wise-uranium.org/rcfdu.html
2. Physics 373 manual writers, "Electronics set-up diagram", Ohio University, Department of Physics and Astronomy, Athens, OH, 45701, USA
3. http://instructor:physics.lsa.umich.edu/advlabs/Alpha_Spectroscopy /alpha spec .pdf
4. http://www.nndc.bnl.gov/nudat2/index_dec.jsp
5. http://www.academia.edu/4191151/Resolution_and_Efficiency_of_High_Purity_ Germanium_Detector
6. http://www.osti.gov/scitech/servlets/purl/310887/
7. Bopp C. J., Lundstrom C. C., Johnson T. M. and Glessner J. J. G. (2009) Variations in 238U/235U in uranium ore deposits: isotopic signatures of the U reduction process? Geology 37, 611614.